THE ALL-IN-ONE SEWER NETWORK MONITORING PLATFORM

Your sewer is telling you two things. Most platforms only listen to one.

Level tells you the water got higher. Velocity tells you why. EDS FlowSense reads both, through Detectronic LIDoTT instruments. It takes them all the way from the chamber to the capital works paper, and nobody re-keys a number in between. Monitoring, analytics, engineering and reporting: one platform, one record, one subscription.

FLOWSENSE PLATFORM DETECTRONIC LIDOTT SENSING

COMPARED WITH ADS, STORMHARVESTER AND KALLIPR PUBLISHED RATE CARD INSIDE

OCTOBER 2026

THE GAP YOU ARE ACTUALLY BUYING AGAINST

A logger portal ends exactly where the engineering starts.

Every monitoring vendor will sell you a screen with a line on it. The line goes up when it rains, and you already knew it rained. What the line will not tell you is how much of that peak was rainfall derived infiltration rather than sewage. It cannot say which catchment the water entered through, or whether the reach is silting up. It cannot say what it would cost per kilolitre to take it out, or whether the number will hold up when your regulator asks how you calculated it.

So the work leaves the platform. It goes into spreadsheets, then into a consultant's engagement, then into a report that arrives three months after the data did. Every handoff re-keys numbers, and every re-key is a place the audit trail breaks.

FlowSense closes that gap by covering the whole chain in one system. Not a portal with analytics bolted on. One continuous path from the instrument in the chamber to the recommendation on the board paper, with every figure traceable back to the reading that produced it.

STAGE 01

Sense

Detectronic LIDoTT instruments in the chamber. Depth to plus or minus 5 mm, velocity where it matters, pressure backup when the radar goes under.

STAGE 02

Receive

Readings keep the logger's own field timestamp. Synthetic snapshots are rejected at the door. A failed batch rolls back rather than half landing.

STAGE 03

Analyse

Rainfall derived infiltration and inflow (RDII) quantified in kilolitres, dry weather flow profiled, blockages named by mechanism, infiltration painted onto the real gravity network.

STAGE 04

Decide

Cost per kilolitre of infiltration removed, net present value (NPV) and benefit cost ratio (BCR), licence compliance, seven report templates under your own branding.

WHY THE INSTRUMENT CHOICE DECIDES THE ANALYSIS

A level only network cannot tell a blockage from a busy Tuesday.

This is the single most important technical point in this document, so it is worth stating plainly. When the overnight minimum in a gravity sewer starts creeping upward week after week, there are two completely different explanations. The pipe is restricting. Or more water is arriving, from a new connection, a growing catchment, or an infiltration source upstream.

On level alone those two look identical. Both lift the trough. Both damp the daily swing. A platform reading only depth has to guess, and the cost of guessing wrong is a jetting crew sent to a pipe that is perfectly clear.

Velocity settles it, because the two cases separate cleanly in the hydraulics. Discharge in a part full pipe is **Q = A × V**, area times velocity. Under Manning, the velocity a given reach can sustain at a given depth is fixed by its slope and roughness. So depth and velocity trace a characteristic curve for that pipe. A rise in load moves the site

along that curve, with higher depth and higher velocity together. A restriction pushes the site *off* it: higher depth at lower velocity, because the effective roughness has risen or the cross section has shrunk.

FlowSense measures exactly that displacement. **Conveyance at matched depth** sorts the recent depth to velocity readings into depth bands and compares velocity in each band with the site's own baseline. Holding depth constant removes the effect of what is arriving through the pipe and isolates the pipe itself. This is the classical scattergraph diagnostic and, in our experience, the earliest of the eight signatures to move.

The same pattern tells you more. A curve that fits a raised invert, with an otherwise normal profile, indicates **silt**. A vertical excursion at constant velocity indicates **surcharge**. Depressed velocity across the whole depth range, with no change in the site's own inflow, indicates a **downstream restriction or backwater**. A jump between two otherwise stable clouds indicates **debris or partial collapse**, which is a CCTV job rather than a jetting job. Where the hydraulics contradict a blockage, FlowSense lowers the score instead of merely failing to raise it. Evidence against a blockage is not the same as an absence of evidence for one.

> The verdict that saves the most money is the one a level only product can never reach: increased load, not an obstruction. No jetting indicated.
>
> FLOWSENSE BLOCKAGE WATCH, MECHANISM CLASSIFICATION

INSTRUMENTATION

What Detectronic puts in the chamber.

Three instruments cover the range from a cheap alarm point to a certified flow monitor, and every one of them feeds the same platform.

Two details on the spec sheets below matter more than they look. **The pressure backup on the Sensor R** means a surcharge that submerges the radar is still measured, just when your network is doing the thing you most need evidence of. A radar only device tops out and the event is lost. **ATEX and IECEx Zone 0** on the Alarm means it is certified for the most hazardous classification of explosive atmosphere, which is what a sewer chamber actually is.

ALARM POINT

LIDoTT Alarm

Sensing	Pulsed coherent radar
Range	8.4 m
Accuracy	±5 mm
Beam angle	5°
Battery	Up to 7 years
Ingress	IP68 / NEMA6P
Hazardous area	ATEX, IECEx Zone 0
Comms	NB-IoT, Cat M1/M2, 2G
Sampling	5 or 15 min
Alarm states	3, reconfigurable OTA

LEVEL MONITORING

LIDoTT Sensor R

Primary sensing	Pulsed coherent radar
Radar range	7 m at ±5 mm
Backup sensing	Silicon diaphragm pressure
Pressure range	10 m at ±0.2% FS
Temperature	±1.5 °C
Battery	Up to 7 years, replaceable
Logging	2, 5, 10 or 15 min
Transmit	5 min to 24 h
Integration	API to any platform

FLOW MONITORING

LIDoTT EDM Sensor M

Sensing	Area velocity plus radar
Measures	Depth and velocity
Certification	MCERTS
Enables	True flow, not inferred
Unlocks	Scattergraph QA
Unlocks	Conveyance at depth
Unlocks	Mass balance pairing
Unlocks	Silt and surcharge detection

SENSING, SIDE BY SIDE

CAPABILITY	DETECTRONIC LIDOTT	ADS TRITON+ AND ECHO	KALLIPR CAPTIS AND SPECTRA	STORMHARVESTER
Hardware supplied	Available from EDS, full LIDoTT range, purchased separately	Yes, TRITON+ flow and ECHO level range	Yes, Captis and Spectra range	No. Analytics only, you supply sensors
Level accuracy	±5 mm to 8.4 m	Quad redundant ultrasonic and pressure depth, figure not published	Non contact radar, figure not published	Depends on your fleet
Measures velocity	Yes, area velocity on EDM Sensor M	Yes, surface and surcharge continuous wave velocity	Not published in the sewer level range	Not supplied
Surcharge above radar range	Captured, pressure sensor takes over to 10 m	Captured, surcharge velocity and pressure depth	Radar range limit	Depends on your fleet
Zone 0 hazardous area certification	ATEX and IECEx Zone 0	ATEX and IECEx Zone 0	Not published	Not applicable
Flow metering certification	MCERTS on the EDM Sensor M	Not published	Not published	Not applicable
Battery life	Up to 7 years	Not published	5 to 10 years by transmission rate	Not applicable

CAPABILITY	DETECTRONIC LIDOTT	ADS TRITON+ AND ECHO	KALLIPR CAPTIS AND SPECTRA	STORMHARVESTER
Install without confined space entry	Yes on the Alarm, self contained unit	ECHO yes under the lid, TRITON+ sensor mounts in pipe	Yes, 10 to 15 minutes claimed	Not applicable

SOFTWARE

Where the three platforms actually stop.

Kallipr Kloud is a fleet and visualisation layer. StormHarvester is an alerting layer. ADS is the one competitor built on the same shape as FlowSense, so the comparison below is the honest one. On operations and hydraulic QA they are excellent, and on several rows they match us. They stop where the engineering turns into a funding decision. FlowSense puts the operations picture, the analytics workbench and the engineering suite in one system.

CAPABILITY	EDS FLOWSENSE	ADS PRISM AND SLIICER	KALLIPR KLOUD	STORMHARVESTER
OPERATIONS				
Live fleet status and mapping	Yes, plus battery, comms and data age health, and every site judged against its own usual pattern	Yes, with ArcGIS Server and GeoServer layers, monitor diagnostics and battery voltage	Yes, Kloud Fleet and Insights	Yes, real time dashboards
Threshold and no data alarms	Yes, email and SMS on state change, per rule recipients, pump condition alarm per finding	Yes, Standard and ANSR alarms, SMS	Yes, with edge filtering	Yes, ML derived alerts
Blockage early warning	Yes, eight signatures including velocity and conveyance	Yes, ECHO early warning of preventable blockages	Pattern alerts	Yes, ML on level plus hyperlocal rainfall
Names the likely mechanism	Yes, silt, sudden obstruction, downstream restriction, or increased load	Not published	Not published	Not published
Backtest against your own overflow register	Yes, one click, no hindsight, non assessable events excluded	Not published	Not published	Published case study results
ANALYTICS				
Inflow and infiltration detection	Yes	Yes, SLiiCER	Yes	Yes

CAPABILITY	EDS FLOWSENSE	ADS PRISM AND SLIICER	KALLIPR KLOUD	STORMHARVESTER
RDII quantified in kilolitres	Yes, peak, volume, percent over DWF, lag time, hydrograph	Yes, storm by storm and system wide, with depth duration frequency	Not published	Not published
Dry weather flow diurnal profiling	Yes, learned per site by day and time	Yes, gross and net wet and dry weather parameters	Not published	Not published
Depth to velocity scattergraph QA	Yes, detects silt, surcharge, backwater and meter fault	Yes, best fit and Stevens Schutzbach curves	Not published	Not published
Upstream to downstream mass balance	Yes, unaccounted volume and error percentage	Flow Balance report	Not published	Not published
ENGINEERING				
Infiltration heat map on the real gravity network	Yes, six selectable methods with kriging confidence overlay	Not offered	Not offered	Not offered
RTK unit hydrograph calibration	Yes, SWMM compatible three component, EPA/600/R-15/162A	RDII published, calibration method not stated	Not offered	Not offered
Pump station volumetric testing	Yes, arriving flow and each pump's delivery against its rating from the wet well trace, plus early warning on output and remaining useful life as a range	Not published	Not offered	Pump alerting
Overflow forecast from a design storm	Yes, predicted hydrograph and capacity exceedance, plus a 72 hour rainfall outlook from ECMWF with its ensemble odds, scored against what fell	Not published	Not offered	Spill analysis
Dynamic wave model of the whole network	Yes, built from your GIS, AR&R 2019 design storms, pump and gate control rules, runs exported to Excel, GeoJSON and SWMM	Not offered	Not offered	Not offered
Capital appraisal, cost per kL of I/I removed	Yes, discounted, with NPV and BCR, ranked	Not offered, delivered as an SSES engagement	Not offered	Not offered
Licence compliance evaluation and SSO register	Yes, instantaneous, daily mean and daily max bases	Fixed report suite, register not published	EPA documentation support	Not published

CAPABILITY	EDS FLOWSENSE	ADS PRISM AND SLIICER	KALLIPR KLOUD	STORMHARVESTER
OWNERSHIP AND CONTROL				
Reads loggers from other vendors	Yes, vendor neutral: three vendor platforms live in production, API and reviewed CSV import for the rest of the market	Partly, USGS feeds and third party FTP import	Built around Kallipr devices	Yes, vendor neutral by design
Report suite under your branding	Seven templates, PDF and CSV, section level control, scheduled by email	Fixed report set, Excel and CSV export	Dashboards and export	Dashboards and reporting
Scheduled delivery into your own systems and SCADA	SFTP, FTPS and MQTT, CSV or JSON, watermarked, usable by any modern SCADA	Export only, no scheduled SCADA delivery published	Configurable endpoints to SCADA	Not published
Accepts a feed from your own SCADA	Yes, five ways in: your SCADA posts to us, we subscribe to your MQTT broker, we poll your API or your supplier's cloud (key or OAuth2 client credentials), we collect historian CSVs from your SFTP, or you upload them to an SFTP folder we host	Third party FTP file sweeping	Not published	Not published
You administer your own users and access	Seven roles, per page grants, named sub teams, no ticket required	Yes, Customer Admin role	Fleet user management	Not published
Shows the calculation behind every figure	Yes, methodology notes bound to the code that computes them	Not published	Not published	Machine learning model

INTEROPERABILITY

Bring the loggers you already own. Keep the SCADA you already run.

FlowSense is vendor neutral at both ends of the pipe. It works with most remote loggers on the market, whoever's name is on the lid. It also goes both ways with your SCADA. Your control room can feed the platform, and the platform can feed your control room, using methods those systems already speak. **Very few products in this category will accept an inbound SCADA feed at all.**

LOGGERS IN

Most loggers on the market

Vendor telemetry platforms	3 live in production
Detectronic LIDoTT range	Native
Any platform with an API	Scoped in onboarding
Any logger that can export	Staged CSV import
Import safety	Review before commit, 30 day undo

YOUR SCADA IN

Any SCADA that can push, publish or export

HTTP push	Your SCADA posts to us
MQTT subscribe	We connect out to your broker
API poll	Your or your supplier's API, key or OAuth2
File pickup	Historian CSVs on your SFTP or FTPS
File drop	Historian CSVs to an SFTP folder we host
Firewall	Nothing inbound for push or drop
Tag mapping	Reviewed, never auto trusted

OUT TO YOUR SYSTEMS

All modern SCADA systems

MQTT publish	JSON or CSV, your broker
Scheduled CSV	SFTP and FTPS
REST API	Documented, token secured
Delivery integrity	Watermarked, retried on failure
Reports	7 templates, PDF and CSV

The inbound half is the unusual one. Most platforms in this market will send data to a SCADA. Taking a council's own SCADA data in, tag by tag, and holding it to the same validation as the logger feeds, is a different proposition. It stops your control room and your flow platform becoming two separate versions of the network. Pump station telemetry, a treatment inlet, a legacy historian series with ten years in it: map it once, and from then on it charts, reports, models and alarms like any other channel.

"Any SCADA" earns its scope from the mechanisms, not from a compatibility matrix. Ignition, VTScada, Geo SCADA, AVEVA, Citect and Plant SCADA, Wonderware, WinCC and their peers can all do at least one of four things. They can run a script that posts readings to a URL, publish to an MQTT broker, answer an API, or write a CSV export to a folder. Those four give the five paths we accept, because a file can travel either way: we collect it from your server, or your system uploads it to a folder we host.

Posting to us and uploading to us need *nothing opened on your side*. For the other three, we connect to one service you choose to expose, from one address you allowlist. We deliberately do not offer direct OPC UA, DNP3 or Modbus polling across the internet. Plant protocols should never be exposed publicly, and every system that speaks them also speaks one of the five above. Each connection is still commissioned with your integrator, tag by tag, rather than promised from a datasheet.

Everything arriving that way passes the same gate as everything else. Each reading keeps your instrument's own timestamp or is refused. A timestamp with no time zone is read in the zone you nominated, not guessed. A future dated row from a SCADA server with a drifting clock is rejected, so it cannot make the site look permanently fresh. Duplicates are dropped on channel and timestamp, so re-sending a window or replaying a file is always safe. A tag nobody has mapped yet is quarantined for review, with a sample value attached. Its readings are counted rather than stored, so nothing your SCADA sends can silently invent a channel. Ingest credentials are write only and separate from the read API keys, so a token in a PLC script cannot read a single stored value.

On the logger side the same logic applies. Three logger vendors' telemetry platforms are read in production today, and the Detectronic range is native. Any vendor platform that exposes an API is a scoped integration task in onboarding. The staged CSV import covers every logger that can produce an export, with review before commit and a thirty day undo. **The claim also costs nothing to test.** The free trial runs on a CSV export, so compatibility with your existing fleet is shown on your own data before a dollar changes hands. We recommend the LIDoTT range where new instrumentation is going in. Vendor neutrality protects the instruments you already own.

DEFENSIBILITY

An answer you cannot audit is not an answer.

StormHarvester's engine is machine learning. That is a legitimate and effective approach, and at scale it finds things. But your regulator, your board or your own asset engineer will ask why a particular site was flagged, or how a volume figure was arrived at. A model score is not an explanation. FlowSense takes the opposite position deliberately.

Methodology bound to the code

Every calculated figure carries a "how this is calculated" note, and those notes import their constants from the modules that do the computing. The documentation cannot drift from the arithmetic, because tests fail if it does.

Real field timestamps only

A reading keeps the time the logger recorded it, never the time we collected it. A record with no trustworthy timestamp is skipped rather than guessed at. Synthetic "as of now" readings, produced for stale sites, are rejected on arrival. Left in, they look like healthy data and quietly hide an outage.

Volumes computed the way your spreadsheet does

Daily volume is the average flow across the local day multiplied by 86,400 seconds, summed, with partial edge days prorated. It matches how water authorities calculate it themselves. It is the first thing an engineer checks, so we say it out loud.

Uncertainty drawn on the map

The infiltration heat map carries a zero to one confidence overlay, so you can see not only the estimate but how much of it is interpolation. Estimated upstream flows are labelled as estimates, not measurements, everywhere they appear.

Rain excluded before anything is judged

Rain produces every blockage signature more strongly than a blockage does. Blockage Watch therefore compares dry days only, treating a day as wet if it or the day before saw meaningful rain. A site that only lifts after rain scores zero and says so.

It refuses to guess

A fortnight too wet to assess is reported as "not enough dry weather", not as "normal". A channel with a constant or negative baseline is flagged as sensor suspect and routed to data quality rather than to a jetting crew. On the first production run that alone separated one genuine finding from eight instrument artefacts, which would otherwise have outranked it.

THE MODULE THE REST OF THE FIELD BILLS AS AN ENGAGEMENT

A consultancy engagement, productised.

The Engineering Workspace is where FlowSense stops being a monitoring product. Fifteen project based modules sit on a GIS canvas carrying your pipes, nodes, pump stations, overflow structures and catchments. You currently commission, wait for and pay for the infiltration study as a report. Here it is a tool you run yourself, against live data, as often as you like.

15
modelling modules, project and scenario based

6
selectable heat map methods, plus a kriging engine

8
blockage signatures across level and velocity, mechanism named

4
formats a finished model run exports to: Excel, GeoJSON, CSV and SWMM

5
inbound paths for your own SCADA, two with nothing to open on your side

7
report templates, operations through to board level

2×
daily Blockage Watch runs across the network

7
access roles with per page grants you control

The heat map deserves particular attention. Most products interpolate infiltration severity in a straight line between gauges. That is geometrically convenient and hydraulically meaningless, because sewage does not travel in straight lines. It travels down pipes. FlowSense paints severity onto the actual gravity network, by catchment and flow direction, following what drains to which meter. You then choose the method that suits your evidence: network attribution, RDII event analysis, hydraulic capacity, depth and velocity condition, upstream to downstream mass balance, or a calibrated RTK model, which fits a rainfall response to each catchment.

Underneath the workspaces sits a full network model. The Hydraulics workspace runs a dynamic wave solver, a full hydraulic simulation, over a network prepared from your own GIS. Friction is set per pipe by Manning or Colebrook-White. Design storms are built to AR&R 2019, and pump and gate control rules cover the assets that actively manage flow. Every run reports its own mass continuity check, so a result that lost water somewhere says so instead of presenting a clean hydrograph. Every run also carries a print ready report of what was simulated, under what assumptions, with what outcome. A finished run leaves the platform as an Excel workbook, GeoJSON,

CSV with its audit trail in the header, or an EPA SWMM input file. The model you build here stays portable to the tools your consultants already use, rather than becoming another form of lock in.

And it ends where your finance team needs it to end. The Capital Plan workspace ranks interventions by **discounted cost per kilolitre of infiltration removed**, with NPV and benefit cost ratio. That is the number that gets a project funded, and it comes out of the same chain that started at the sensor.

ANALYTICAL BASIS

Every method, and the standard it is worked to.

Set out here because this is the page your technical evaluator will turn to first, and because a vendor unwilling to publish its methods is telling you something.

> FlowSense is built to the published standards below. The methods, acceptance bands and thresholds are taken from the standards themselves, and each is wired to the module that implements it. The test suite fails if the software stops agreeing with them.
>
> CONFORMANCE HELD BY TESTS, NOT ASSERTED ONCE

CIWEM UDG Code of Practice

Hydraulic Modelling of Sewer Systems, 2017. Chartered Institution of Water and Environmental Management, Urban Drainage Group. The per storm acceptance bands every calibration and validation is scored against, and the dry weather flow pattern split by day type.

WEF MOP FD-6

Existing Sewer Evaluation and Rehabilitation, WPCF Manual of Practice No. 36. Water Environment Federation. Dry weather flow as the daily sanitary pattern plus groundwater infiltration, from qualifying dry days only.

QARTOD

Manual for Real-Time Quality Control of In-Situ Data. US Integrated Ocean Observing System. The quality control tests and flag vocabulary that judge every sample.

WSA 02

Sewerage Code of Australia. Water Services Association of Australia. Self cleansing velocity, level of service checks, and the friction law that capacity is checked against.

AS/NZS 3500.3

Plumbing and drainage, stormwater drainage. Standards Australia and Standards New Zealand. The Rational Method for sub-catchments with no gauge, and the minimum velocity band on scattergraphs.

AR&R 2019, Book 2

Australian Rainfall and Runoff. Geoscience Australia and Engineers Australia. Areal reduction factors, and AR&R temporal patterns run exactly as supplied.

NSW Liquid Trade Waste Guidelines

Liquid Trade Waste Management Guidelines, 2021. NSW Department of Planning and Environment, Water. The charging formulas the trade waste module works to: annual fee, usage, excess mass with the escalating BOD5 rate, and the pH charge. Each council sets its own fees.

EPA SWMM

Reference Manual Volume II Hydraulics, and EPA/600/R-15/162A for RDII. United States Environmental Protection Agency. The dynamic wave routing the network solver belongs to, and the RTK unit hydrograph method.

ISO 13381-1

Condition monitoring and diagnostics of machines, prognostics. The convention for stating a pump's remaining useful life: a range from the trend in its measured output, projected to a stated limit, and never further ahead than the record reaches back.

ISO 8601 and WCAG 2.1 AA

Every timestamp crossing the platform boundary, and the contrast minima the interface colour system is built to, in both light and dark.

In the product this is one page. **How it is calculated**, under Reference, lists every standard alongside the calculations that apply it. Each calculation opens to its inputs, its steps, its thresholds and the reason each threshold has the value it has. The page is generated from the running software, so a standard cannot appear on it unless something in the product uses it. Ask us to walk you through any of it against the code.

ANALYSIS	METHOD AS IMPLEMENTED	BASIS OR REFERENCE
Flow volume	Mean flow per local day times 86,400 s, summed across the period, partial edge days prorated by window overlap	Matches water authority spreadsheet practice, reported in kL
Hydraulic QA	Observed depth to velocity cloud fitted against the theoretical curve, with silted invert fitting	Manning's equation, scattergraph analysis
Dry weather flow	Daily profile learned per site by day of week and time of day, plus a night baseline	WEF MOP FD-6 and the CIWEM UDG Code of Practice. Site is its own control, no regional assumption
RDII quantification	Hydrograph separation against the learned DWF baseline, reporting peak RDII, volume, percent over DWF and lag time	Rainfall derived infiltration and inflow
Unit hydrograph model	Three component R, T, K parameters calibrated per catchment, fitted on hydrograph shape, event volume and event peak together and validated on an independent event set. Nash-Sutcliffe and Kling-Gupta both reported, with a 95% confidence interval on every fitted parameter	US EPA/600/R-15/162A, SWMM compatible. Kling-Gupta after Gupta et al. 2009
Model verification	Every storm scored separately: simulated peak within 15% below to 25% above observed, event volume within 10% below to 20% above, peak timing reported against a one hour tolerance. Verdict: verified, partially verified, not verified or insufficient events, with three assessable storms the minimum for verified	CIWEM Urban Drainage Group Code of Practice for the hydraulic modelling of urban drainage systems, 2017

ANALYSIS	METHOD AS IMPLEMENTED	BASIS OR REFERENCE
Pipe capacity	Full bore and part full capacity from the pipe's own geometry, by Manning's equation or by Colebrook-White with an equivalent sand roughness. Any pipe that never reaches self-cleansing velocity is flagged	WSA 02 Sewerage Code of Australia practice, sliming assumed rather than clean new pipe
Network hydraulic model	Dynamic wave routing over the prepared network, friction per pipe by Manning or Colebrook-White, pump and gate control rules, and a mass continuity check reported on every run	EPA SWMM Reference Manual Volume II hydraulics, results exchangeable as a SWMM input file
Design rainfall	IFD depths and temporal patterns for scenario and overflow forecasting	Australian Rainfall and Runoff 2019
Ungauged sub-catchments	Rational Method inflow estimation where no meter exists	AS/NZS 3500.3 and AR&R, presented as estimate not measurement
Channel routing	Muskingum-Cunge routing for multi-site concurrent simulation	Standard hydrologic routing
Spatial interpolation	Gaussian process regression with a zero to one confidence field rendered as an opacity overlay	Kriging, uncertainty surfaced not hidden
Network attribution	Severity painted along the gravity network by catchment and flow direction, drains-to-meter tracing at 5 m snap tolerance	Topological, not straight line interpolation
Derived flow channels	Area multiplied by velocity, or Manning's equation where the logger reports depth only	Appears as a first class channel throughout
Data quality tests	QARTOD suite: gross range, climatology, Hampel filter for local spikes, rate of change limits, flat line and stuck at zero detection, attenuated signal, cross channel regression residuals	US IOOS QARTOD test battery and flag vocabulary. Gap fills remain export only, never written back
Rainfall record	Modelled rain at each site's own grid cell, ECMWF IFS at 9 km via Open-Meteo, filed in the hour it fell and refreshed six hourly. A gauge you upload or carry on telemetry takes precedence. A missing reading is never treated as zero millimetres. The 72 hour outlook adds the odds from ECMWF's 51 member ensemble and is scored two days later against what fell	Absence of record is not absence of rain, and a forecast is checked rather than trusted
Blockage detection	Eight signature fingerprint across level and velocity, dry days only, suspect sensors screened out, mechanism classification	Deterministic and inspectable, not a model score
Pump condition	Each pump's measured output on dry days against its own reference, slow decline and sudden drop flagged, remaining useful life as a range from a Theil-Sen trend, cause ranked rather than diagnosed	ISO 13381-1 convention for the life figure; hydraulic output only, not bearings, seals or windings

PLATFORM ENGINEERING

The integrity work that happens before any analysis runs.

Analytical sophistication is worthless on corrupted inputs. A meaningful share of the engineering in FlowSense sits below the analytics, in how a reading gets into the record. It exists because each item on this list is a failure mode we have met in production telemetry.

Timestamp provenance

Every reading carries the logger's own field time. Collection time is never substituted, and a record without a trustworthy timestamp is skipped rather than stamped with the current clock. Silent clock substitution is the most damaging single defect in telemetry, because the data looks perfect while every downstream aggregate is wrong.

Synthetic snapshot rejection

A quiet site often produces an "as of now" record that looks like a real reading. Validation on arrival identifies and rejects these. Accepted, they mask an outage indefinitely and the no data alarm never fires.

All or nothing writes

A batch that fails part way rolls back, so a partial write never lands. Collection for a site also cannot run twice at once or write twice.

Gap detection and targeted repair

A long outage leaves a gap in the record that never fills itself in. A dedicated recovery tool finds those windows and repairs them, rather than leaving a hole nobody notices until a report is already wrong.

Forwarding watermarks

Scheduled delivery to your systems over SFTP, FTPS or MQTT moves its watermark, the bookmark that records what has been sent, forward only after a fully successful send. A failed delivery is retried, not skipped.

Architecture

FastAPI services against PostgreSQL, a React single page client, an APScheduler job runner and containerised deployment. Sign-in uses JWT with refresh tokens, and credentials are encrypted at rest. WebSocket push falls back to polling. Each module has its own error boundary, so a fault in one workspace cannot blank the application. When a third party GIS or weather service is unavailable, the platform degrades gracefully to cache.

SAID PLAINLY

Where the others are genuinely good, and what we will not claim.

You will hear a version of this from their sales teams too. Better that you hear the honest version from us first. The parts we are careful about are the parts that show we are careful.

On our competitors

ADS ENVIRONMENTAL SERVICES

The most complete competitor in this document, and the one we measure ourselves against. ADS has four decades of flow monitoring practice and is now part of IDEX. It is the origin of much of the depth to velocity scattergraph QA the whole industry now relies on. TRITON+ measures depth and velocity and carries ATEX and IECEx Zone 0. PRISM is a mature operations platform with ArcGIS Server integration, monitor diagnostics and a report suite we respect. SLiiCER quantifies RDII storm by storm and system wide, with depth duration frequency analysis. We do not claim to beat ADS on monitoring practice, and the table above concedes the rows where they match us. Our argument is narrower. ADS publishes the RDII number and then sells you the study. There is no dynamic wave network model, no infiltration map constrained to your own gravity network, no cost per kilolitre appraisal, and no scheduled delivery back into your SCADA. We can find no Australian field service operation, which in practice is the difference between a comparison and a procurement.

KALLIPR

Kallipr is an Australian company, founded in Brisbane in 2017 and manufacturing here. It makes genuinely good hardware. The Captis and Spectra devices are rugged and install fast without confined space entry, and their battery life is competitive with anything on the market. If your requirement is a large fleet of level points managed cleanly, Kloud Fleet does that well. Our argument is not that their loggers are poor. Kloud is a device and visualisation platform, so the engineering work still has to happen somewhere else. And the fleet is built around their own devices. The second difference is focus. Kallipr sells into water, rail, mining and environmental monitoring, so their platform has to generalise across four industries, and their recent expansion has been toward the North American market. EDS builds for Australian wastewater networks and nothing else. That is the whole reason the engineering suite goes as deep as it does.

STORMHARVESTER

StormHarvester runs the largest proactive sewer monitoring analytics deployment we know of. It reports a hit rate of roughly 70 percent on predicted blockages, with real results at Anglian Water, Wessex Water and now Barwon Water and Watercare. Their hyperlocal rainfall correlation is strong, and they are vendor neutral, which is a real virtue. Our argument is that

they are an alerting layer on level data. They will tell you a site is behaving oddly. They will not calibrate your RTK model, rank your capital works by cost per kilolitre, or evaluate your licence compliance.

THE WIDER FIELD

Three other groups compete for the same budget, and it is fairer to name them than to pretend the market is three companies wide. The first is the Australian level vendors: Taggle with its own low power wide area network and Aqualus Vision, and Iota with Level Lotic behind South East Water. They build good hardware and stop at the alarm. The second is the software only analytics firms, iFactory here and infinitii ai in Canada. They pitch the same separation of dry weather flow, groundwater infiltration and rainfall driven inflow that we do, with no instrument of their own and nothing to sell you in a chamber. The third is the enterprise platforms: Autodesk Info360 Insight alongside InfoWorks ICM, plus Xylem, Idrica and Bentley. They will do everything on this page and a great deal more, at utility scale. Their capacity based subscriptions and per seat modelling licences are ones a shire council is never going to sign. We are aimed squarely at the authority that is too small for the enterprise platforms and too serious for an alarm.

On ourselves

VELOCITY COVERAGE

The velocity signatures need velocity instrumentation. On our production network in August 2026, 37 of 91 sites were instrumented for velocity. Six of those had probes reporting zero, and were flagged for repair rather than misread as a blockage. Sites without usable velocity data are assessed on level alone, at lower confidence. The recommendation that follows is to instrument the reaches that matter with the EDM Sensor M, not to instrument everything.

BLOCKAGE WATCH LIMITS

It is a statistical early warning from a sensor's history against itself. It says the pipe reaching that sensor is behaving differently for a reason rain does not explain. It does not locate the obstruction and cannot see one downstream of the meter. It needs roughly two and a half months of history plus enough dry days. We quote precision only once cases have been closed with recorded outcomes on your own network, never as a headline figure borrowed from someone else's.

SCADA CONNECTIONS

The five inbound paths are built and running. Between them they cover the systems named earlier. We have not tested FlowSense against every vendor's build in a lab, and we will not pretend otherwise. What we commit to is the mechanism. If your SCADA can post to a URL, publish to an MQTT broker, answer an API or export a CSV to a folder, we take its data. We

commission the connection with your integrator, tag by tag, before anything counts as live. Direct OPC UA, DNP3 and Modbus polling over the internet are not offered. That is a deliberate security position, not a roadmap item.

GIS INTEGRATION

The network layer is live against council ArcGIS services, currently proven on one regional council's public gravity main layer of roughly 28,000 segments across ten schemes. For a new authority this is an integration task, not an out of the box switch.

THE NETWORK MODEL'S SCOPE

The dynamic wave solver models your gravity network in one dimension, with pumps, gates and control rules driven by the same telemetry as everything else. It does not do 2D overland flood mapping, water quality or sediment transport, which remain work for a dedicated modelling suite. That is why a finished run exports as an EPA SWMM input file. Where a study needs those layers, the network you prepared and the model you calibrated leave with you, instead of being rebuilt from scratch.

OVERFLOW FORECASTING

As good as the rainfall forecast fed into it. Present it as a risk ranking tool, which is what it is, not as a guarantee about a specific night.

LOCAL

Built here, for the way Australian authorities actually report.

FlowSense handles local time throughout. That sounds trivial until a daily volume total lands in the wrong day and a monthly compliance figure moves. Days are grouped in local time and dates are entered in the format your team already uses. The header clock names your zone the way a control room says it: AEST, Brisbane, or AEDT, Sydney. Design rainfall follows AR&R 2019.

EDS is an Australian business. Support comes from people who have stood in the chambers your instruments are going into. The platform is under continuous development against real client networks. What those networks turn out to need shapes the roadmap, not a release cycle set on another continent.

THE ASSESSMENT

On the evidence set out here, FlowSense is the stronger option. Three reasons.

Not because it has more features. Because of three specific things that determine whether a monitoring investment produces defensible engineering outcomes or another dashboard.

01 It measures the quantity that resolves the ambiguity

Blockage and load are indistinguishable on level. Detectronic's area velocity instruments supply the second quantity. Conveyance at matched depth then turns it into a diagnosis of the pipe rather than of its catchment. Kallipr's sewer level range publishes no velocity measurement, and StormHarvester supplies no instrument at all. Neither can conclude that a rise is load rather than obstruction. That single distinction is the difference between a targeted jetting programme and a crew sent to a clear pipe.

02 It carries the work to the funding decision

Kloud stops at visualisation. StormHarvester stops at the alert. ADS stops at the report it writes for you, on its own engagement clock. FlowSense carries on through RDII quantification, RTK calibration to EPA/600/R-15/162A, network constrained infiltration mapping and licence compliance. It finishes at discounted cost per kilolitre of infiltration removed, with NPV and BCR. The analysis that now arrives as a consultant's report three months after the data is available continuously, from the same chain. There is no re-keying and no break in the audit trail.

03 Its outputs survive scrutiny

Every figure shows its method, and those method notes take their constants from the code that performs the calculation, so the documentation cannot drift from the arithmetic. Uncertainty is drawn, not hidden. Estimates are labelled as estimates. Rain is excluded before any dry weather judgement is made, and the detector declines to report where evidence is insufficient. When a regulator or a board asks how a number was reached, there is an answer, and it is the same every time.

The honest qualification, stated above and repeated here, is twofold. Velocity signatures require velocity instrumentation, and GIS integration for a new authority is a scoped task rather than a switch. Both are addressed in deployment planning, and neither alters the assessment.

PRICING

One subscription. Every capability. Priced by the site.

Published here, because a price you can only discover through a sales cycle is itself a cost. There are no editions, no modules and no per user fees. Every capability in this document is in the one subscription, and the only number that moves is how many sites you monitor. Two things sit outside it. The instruments are a separate one off purchase. The optional EDS service runs a calibration with you, rather than you running it yourself.

SUBSCRIPTION RATES, AUD EXCLUDING GST

BAND	PER SITE PER MONTH	PER SITE PER YEAR
First 20 sites	$30	$360
Sites 21 and beyond	$20	$240

Bands are cumulative: a fifty site network pays the first rate on twenty sites and the second on thirty. Growing the network never re-prices the sites you already have. A site is a monitored location with at least one active channel, and archived sites cost nothing. Billing is annual, on twelve month terms. You can cancel at renewal, and your data exports with you in full. The only pass through is SMS alerting at carrier cost.

WHAT THAT MEANS FOR A REAL NETWORK

NETWORK	SITES	PER YEAR	PER MONTH	BLENDED PER SITE PER MONTH
Small shire	5	$1,800	$150	$30
Regional council	20	$7,200	$600	$30
Mid size council	50	$14,400	$1,200	$24
Large regional network	90	$24,000	$2,000	$22
Bulk utility	300	$74,400	$6,200	$21

The bands are set deliberately against procurement reality. For most Queensland councils a fifty site network sits under the $15,000 small contractual arrangement threshold. That means a single quote and a delegated signature rather than a procurement round. Even a three hundred site utility sits far below any tender threshold in the country.

$0
per user, per report, per export. Sites are the only meter.

100%
of platform capability in every subscription, no editions

12 mo
terms, cancel at renewal, no lock in contract

90 days
free trial on your own uploaded history, no hardware needed

WHERE THAT SITS IN THE MARKET

OFFERING	WHAT IT IS	PRICE AS PUBLISHED OR CONTRACTED	APPROXIMATE AUD
EDS FlowSense	Whole platform, analytics and engineering suite, unlimited users	$20 to $30 per site per month, all inclusive	$22 blended at 90 sites
SOFTWARE ONLY ANALYTICS, PER METER PER MONTH			
H2Ometrics Advanced (US)	Sewer flow and infiltration analytics SaaS, capped at 10 users	US$85	about A$130
H2Ometrics Basic (US)	Data hosting, scatter plots and rainfall tools only	US$20	about A$31
ADS PRISM software access (US)	Data viewing software line in a live municipal contract	US$25	about A$39
MONITORING SERVICE INCLUDING HARDWARE, PER SITE PER MONTH			
SmartCover, City of Beaumont CA	Eight level monitors, three year contracted programme	US$185	about A$285
ADS full service, City of Cartersville GA	Flow monitors, field service, communications and software	US$1,022	about A$1,570
MODELLING SOFTWARE, PER SEAT PER YEAR			
Autodesk InfoWorks ICM Sewer	Sewer network modelling, one user	US$5,500	about A$8,500
Autodesk InfoWorks ICM Ultimate	Full integrated catchment modelling, one user	from US$18,000	about A$27,700
Bentley SewerGEMS	Sewer modelling, Virtuosity subscription, one user	about US$7,600	about A$11,700
DHI MIKE+ Collection Systems	Collection system modelling, one user	€9,000	about A$15,300
THE PLATFORMS COMPARED IN THIS DOCUMENT			
Kallipr Kloud	Device fleet and visualisation platform	Not published, quotation only	
StormHarvester	Machine learning alerting on level data	Not published, quotation only	

> A 90 site network, every module, unlimited users: $24,000 a year. Less than one seat of InfoWorks ICM Ultimate, the modelling software it can calibrate.
>
> RATE CARD, SEPTEMBER 2026

That positioning is deliberate. The nearest published software comparable charges roughly six times this rate card for its full tier, and caps the users. The consultancy study this platform productises is a one off engagement that costs more than several years of the whole subscription. The intent is not to be slightly cheaper. It is to make continuous network analytics an unremarkable line item that every authority can carry, on every network, permanently.

GETTING STARTED

STEP	WHAT IT COVERS	PRICE
Free trial	90 days, the whole platform, running on your own uploaded flow history. Upload last winter's CSV exports and see your infiltration ranking the same afternoon. No hardware required.	$0
Assisted onboarding	GIS network import, site and channel configuration, alarm setup, commissioning any SCADA connection you want feeding the platform, and training.	$2,500 once, waived at 50 sites and above
Optional EDS Guided Calibration Service	The calibration capability itself is included in every subscription and you can run it yourself. This is the optional service where EDS does the work with you. It covers RTK model calibration and CIWEM standard verification for one catchment, delivered in the platform where you can re-run it, not as a PDF.	$7,500 per catchment

NEXT STEP

See it running against a real network, not a slide.

We will demonstrate FlowSense live, including the blockage backtest replayed against a real overflow register. We will show you the working behind every number on the screen. Bring your hardest question about how a figure was calculated.

Environmental Data Services

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Competitor capabilities are characterised from each vendor's public material as at September 2026. Cells reading "not published" mean the capability was not documented in that material and should not be read as confirmation it is absent. Where a competitor demonstrates a capability we have marked this way, we will correct it. Vendor names and product names are the trademarks of their respective owners.

Comparator pricing is as published by the vendor or as recorded in public municipal contract documents at August 2026, converted at approximately A$1.54 per US dollar and A$1.70 per euro and rounded. Those figures are indicative context, not quotations. EDS rates on the pricing page are the current founding rate card in Australian dollars excluding GST, and they are the rates we will quote.

DOCUMENT CONTROL

DOCUMENT	REFERENCE	REVISION
EDS FlowSense competitive brochure	EDS-FS-CMP-01	2.5, issued 12 October 2026
SUPERSEDES	**STATUS**	**ISSUED BY**
Revision 2.4, 11 October 2026	Issued for client use	Environmental Data Services, ABN 31 729 813 151